



Thesis Proposal

AUTONOMOUS VEHICLE NAVIGATION SYSTEM USING MACHINE LEARNING AND GRAPH BASED SLAM

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PREFACE

Praise and gratitude to the presence of Allah SWT. for all His grace, the author was able to complete this research with the title **AUTONOMOUS VEHICLE NAVIGATION SYSTEM USING MACHINE LEARNING AND GRAPH BASED SLAM**. This research was written as a requirement for completing a master's degree at the Department of Electrical Engineering, Institut Teknologi Sepuluh Nopember.

The author also expresses great thanks to:

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ABSTRACT

Icar is an autonomous vehicle developed by Institut Teknologi Sepuluh Nopember (ITS) to autonomously navigate the campus environment. Its navigation system relies on pose estimation using sensor fusion between odometry and the Global Navigation Satellite System (GNSS). However, GNSS signals are prone to degradation in environments with obstacles such as tall buildings and trees, while odometry suffers from inaccuracies caused by wheel slippage, encoder drift, and inertial sensor errors. These limitations result in unreliable localization, potentially causing Icar to deviate from its designated trajectory.

To address these issues, this research proposes an enhanced navigation system by integrating a camera and LIDAR. Road detection is performed using a semantic segmentation model based on deep learning, enabling Icar to distinguish road surfaces from non-road areas. Pose refinement is carried out using graph-based optimization within a SLAM framework, incorporating data from the depth camera and LIDAR processed via the Iterative Closest Point (ICP) algorithm. The proposed system improves localization robustness and offers a GNSS-independent alternative for safe and accurate autonomous navigation. The final system is expected to reduce dependency on GNSS and increase Icar's reliability in semi-structured or dynamic environments.

Keyword: Autonomous vehicle, Icar, GNSS, Odometry, Pose Estimation, Camera, Semantic Segmentation, SLAM, ICP, Graph-Based Optimization, Deep Learning

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NOMENKLATUR

Symbol / Acronym	Description
x_{waypoint}	X-coordinate of the waypoint from the predefined trajectory
y_{waypoint}	Y-coordinate of the waypoint from the predefined trajectory
x_{position}	Current X-coordinate of the vehicle
y_{position}	Current Y-coordinate of the vehicle
$\theta_{\text{direction}}$	Heading angle toward the waypoint
$\theta_{\text{orientation}}$	Current orientation of the vehicle
v_{target}	Target velocity of the vehicle
v_{max}	Maximum velocity allowed
δ_{target}	Target steering angle based on bicycle model
L	Distance between the front and rear wheels
$D_{\text{lookahead}}$	Lookahead distance for path tracking
GNSS	Global Navigation Satellite System
SLAM	Simultaneous Localization and Mapping
ICP	Iterative Closest Point
IMU	Inertial Measurement Unit
EKF	Extended Kalman Filter
GTSAM	Georgia Tech Smoothing and Mapping
DoF	Degrees of Freedom
RTK	Real-Time Kinematic (high-precision GNSS)
DGNSS	Differential GNSS
ML	Machine Learning
CNN	Convolutional Neural Network
Fast-SCNN	Fast Semantic Segmentation Convolutional Neural Network
ReLU	Rectified Linear Unit
SVD	Singular Value Decomposition
LIDAR	Light Detection and Ranging
PCL	Point Cloud Library
Odometry	Estimation of position based on wheel rotations and IMU data
Pose	A combination of position and orientation in 2D or 3D space
Yaw	Rotation around the vertical (z) axis
Kalman Filter	Recursive algorithm for optimal state estimation
Multipath	GNSS signal distortion caused by reflection or obstruction
Trajectory	A planned path for the robot to follow
Feature Matching	Process of aligning keypoints across frames or sensors

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CHAPTER 1

INTRODUCTION

1.1 Background

Icar is an autonomous vehicle developed by Institut Teknologi Sepuluh Nopember (ITS). Icar is designed to operate automatically and navigate throughout the ITS campus. Its control system relies on pose estimation (i.e., position and orientation) derived from odometry and sensor fusion with the Global Navigation Satellite System (GNSS). The estimated pose is then used to guide Icar toward a predetermined destination.

The pose of Icar is determined through the combination of odometry and GNSS data. Odometry estimates pose based on wheel rotations, augmented by a gyroscope to obtain orientation. GNSS, on the other hand, estimates pose based on satellite signals. These two sources are fused using the Extended Kalman Filter algorithm to achieve higher accuracy in pose estimation [26].

However, the data provided by GNSS is not always fully reliable. Several environmental conditions, such as tall buildings or dense tree canopies, can obstruct satellite signals. This may cause a phenomenon known as multipath, in which satellite signals reach the receiver via multiple indirect paths, resulting in errors in position and orientation estimation [10].

Errors can also arise from the odometry system due to factors such as inaccurate wheel travel distance, incorrect wheel rotation angle calculations, or faulty IMU sensor readings. These inaccuracies may result from differences in wheel diameters, uneven tire pressure, or wheel slippage. Moreover, errors in wheel rotation angle measurements may originate from malfunctioning encoder sensors [2].

When errors occur in pose estimation, Icar may deviate from its intended trajectory. This can lead to safety concerns, such as collisions with obstacles or deviation from designated transportation lanes. Therefore, an additional navigation system is required to enhance pose accuracy and ensure safe

autonomous navigation.

A promising enhancement is the integration of a camera with LIDAR, which provides visual and depth information to detect roads. This road detection process employs a machine learning algorithm—specifically, semantic segmentation—trained on labeled datasets to distinguish between road and non-road areas [21].

Beyond road detection, the camera and LIDAR is also used to refine pose estimation through graph-based optimization. Furthermore, a LIDAR sensor is incorporated and processed using the Iterative Closest Point (ICP) algorithm to further improve localization accuracy. Together, these methods form a Simultaneous Localization and Mapping (SLAM) system [22], which is expected to potentially replace the role of GNSS entirely.

1.2 Formulation of the Problem

The main issue with Icar lies in its navigation system’s heavy dependence on GNSS. When GNSS signal errors occur, or when odometry is affected by slippage or IMU drift, the resulting pose estimation becomes inaccurate. Consequently, Icar may deviate from its intended path, increasing the risk of collisions or deviation from public transportation routes.

1.3 Objective

The objective of this research is to develop a robust navigation system for Icar that is capable of correcting pose estimation errors caused by GNSS inaccuracies or odometry drift. This system incorporates a camera and LIDAR to perform both road detection and pose correction. The goal is to enable Icar to navigate more accurately and reliably in dynamic environments.

1.4 Scope and Limitations

This study focuses exclusively on Icar, an autonomous vehicle developed by ITS. The scope is limited to the development and implementation of a navigation system that utilizes a camera and LIDAR. This research does not cover the electronic or mechanical aspects of Icar’s control system.

1.5 Contribution

The expected contribution of this research is to provide a reference framework for enhancing the accuracy and safety of autonomous vehicle navigation systems. By integrating advanced perception and localization technologies, Icar is expected to achieve improved trajectory tracking and obstacle avoidance, supporting its intended operation in complex urban environments.

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CHAPTER 2

LITERATURE REVIEW

To support this research, several relevant theories are required as references and conceptual foundations. This will help ensure the research is well-directed and aligned with existing scientific knowledge.

2.1 Odometry

Odometry is an algorithm used to estimate how far a robot has traveled. In a 2D pose system, the robot's movement is represented as x, y, yaw . The simplest form of 2D odometry combines motor rotation data with orientation data from an Inertial Measurement Unit (IMU). By applying forward kinematics to this sensor data, the robot's pose can be estimated relative to its initial position [26].

2.2 Bicycle Model

The bicycle model is a simplified mathematical representation of vehicle dynamics. It models a vehicle as a two-wheel system—one at the front and one at the rear—mimicking the dynamics of a bicycle. This model is widely used in vehicle control and trajectory planning due to its simplicity and ability to capture basic vehicle behavior [20]. In robotics and electrical engineering, the bicycle model is commonly used to develop navigation and motion control algorithms for autonomous ground vehicles.

$$\delta_{\text{target}} = \tan^{-1} \left(\frac{2 \cdot L \cdot \sin(\theta_{\text{direction}})}{D_{\text{lookahead}}} \right) \quad (2.1)$$

where δ_{target} is the target steering angle, L is the distance between the front and rear wheels, $\theta_{\text{direction}}$ is the heading angle, and $D_{\text{lookahead}}$ is the lookahead distance.

The main advantage of the bicycle model is its applicability in control and simulation algorithms such as PID, LQR, and MPC. Many autonomous

vehicle systems use it as a foundational model for motion prediction and path following. For instance, it enables calculation of the optimal steering angle required for a vehicle to accurately and stably follow a predefined trajectory [18].

2.3 GNSS

The Global Navigation Satellite System (GNSS) is a satellite-based system that provides highly accurate global information on position, velocity, and time. GNSS encompasses multiple satellite navigation systems, including GPS (USA), GLONASS (Russia), Galileo (EU), and BeiDou (China), which can function independently or in combination [16]. In electrical engineering and robotics, GNSS plays a critical role in navigation and positioning, particularly for mobile robots operating in open environments.

GNSS operates by measuring the time required for satellite signals to reach a receiver. A GNSS receiver can compute a 3D position using signals from at least four satellites. Techniques such as Differential GNSS (DGNS) and Real-Time Kinematic (RTK) improve accuracy by applying corrections from fixed reference stations [8]. These techniques are especially beneficial in applications like precision agriculture, drones, and autonomous vehicles.

GNSS can be integrated with other systems such as Inertial Navigation Systems (INS) to enhance reliability in environments where satellite signals are blocked, such as under tree canopies or between buildings [6]. Integration with sensors like IMUs, LiDARs, and cameras also enhances performance in SLAM systems.

Beyond navigation, GNSS contributes to control systems, trajectory planning, and multi-robot coordination. With ongoing advances in satellite infrastructure and error correction algorithms, GNSS remains a cornerstone for the development of efficient and precise outdoor robotic systems.

2.4 Kalman Filter

The Kalman Filter is a powerful linear estimation algorithm used to predict and update the state of dynamic systems based on noisy sensor

data. Introduced by Rudolf E. Kalman in 1960, it enables accurate real-time estimation of variables such as position and velocity [7]. In robotics and control systems, the Kalman Filter is instrumental in fusing data from multiple sensors such as GNSS, IMUs, and wheel encoders.

The algorithm consists of two stages: prediction and update. During the prediction stage, the system state is estimated using a motion model. In the update stage, the estimate is refined based on new sensor measurements. This iterative process results in increasingly accurate state estimations [24].

2.5 SLAM

Simultaneous Localization and Mapping (SLAM) is a foundational technology in autonomous robotics. It allows a robot to construct a map of an unknown environment while simultaneously estimating its own position within that map. SLAM is particularly valuable in GPS-denied environments and is widely used in autonomous vehicles, drones, and mobile robots [3].

2.6 Graph-Based SLAM

SLAM techniques can be divided into two broad categories: filter-based SLAM (e.g., Extended Kalman Filter SLAM) and graph-based SLAM. In filter-based SLAM, the robot's pose and landmark positions are treated as stochastic variables updated recursively. In contrast, graph-based SLAM builds a graph where nodes represent robot poses and landmarks, and edges represent spatial constraints derived from sensor observations [22].

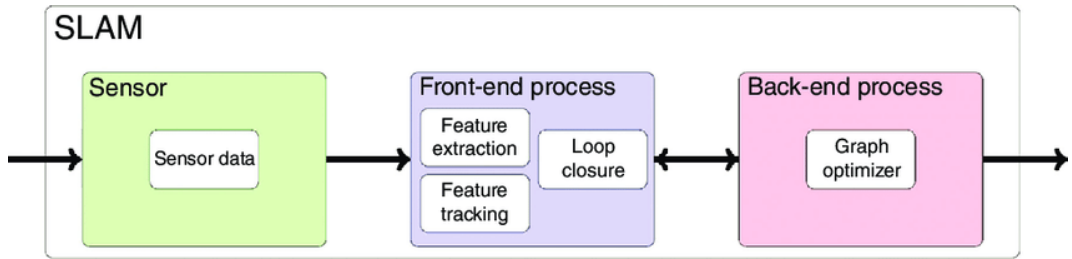


Figure 2.1. Basic Graph-Based SLAM Architecture [25]

There are three main process that can be seen from figure 2.1: The first process is sensor data, it means like acquiring data from sensors such

as cameras, LIDAR, or IMU. The second process is front end process, The front end process is responsible for extracting features from the sensor data and finding a loop closure. The last process is back end process, the back end process is responsible for optimizing the graph to find the best estimate of the robot's trajectory and the map of the environment [25].

The optimization in Graph-based SLAM can be efficiently implemented using frameworks like GTSAM (Georgia Tech Smoothing and Mapping), a C++ library that applies nonlinear optimization over factor graphs. Factor graphs represent probabilistic relationships among variables such as odometry, landmarks, and sensor observations, and allow for efficient optimization of robot trajectories [4].

2.7 RTAB-Map

RTAB-Map (Real-Time Appearance-Based Mapping) is a graph-based SLAM framework designed for real-time applications in robotics. It integrates various sensors, including cameras, LIDARs, GPS, IMUs, and Odometry, to build a map of the environment while simultaneously localizing the robot within that map. RTAB-Map is particularly effective in large-scale environments and can handle loop closures and dynamic changes in the environment [13].

Stereo cameras are widely used in robotics due to their ability to generate real-time depth maps without requiring active sensors like LiDAR. They are especially useful in lightweight and power-efficient robotic platforms such as drones and small autonomous vehicles [11].

2.8 Machine Learning

Machine learning (ML) is a subset of artificial intelligence that focuses on creating algorithms capable of learning from data and improving performance without being explicitly programmed [17]. In robotics and electrical engineering, ML is essential for building intelligent systems that can recognize patterns, make decisions, and adapt to changing environments.

ML methods are typically categorized into three types: supervised learn-

ing, unsupervised learning, and reinforcement learning [5]. Supervised learning uses labeled datasets for tasks like classification and regression. Unsupervised learning deals with unlabeled data for clustering or dimensionality reduction. Reinforcement learning enables systems to learn optimal behaviors through reward-based interactions with the environment.

A key ML technique in robotic perception is semantic segmentation, which classifies each pixel in an image into predefined categories. This technique is widely used for road detection, scene understanding, and navigation.

2.9 Semantic Segmentation

Fast-SCNN (Fast Semantic Segmentation Convolutional Neural Network) was designed for real-time performance on resource-constrained devices, making it ideal for embedded systems like autonomous vehicles. It combines a lightweight encoder-decoder structure with efficient components such as depthwise separable convolutions and a streamlined feature fusion strategy. These design choices enable Fast-SCNN to achieve a good balance between accuracy and speed, ensuring that road segmentation can be performed in real time without requiring GPU-level hardware.

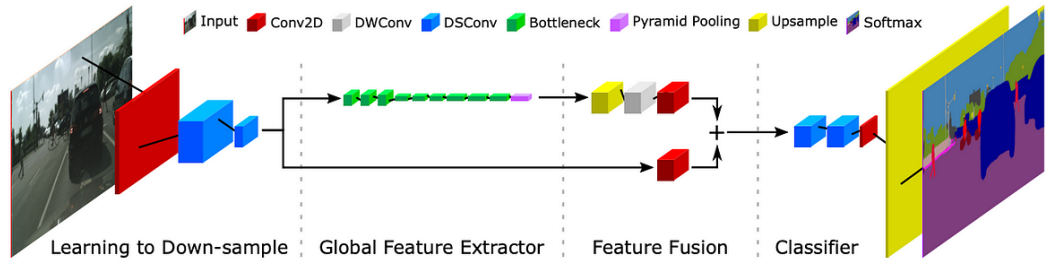


Figure 2.2. Fast SCNN Architecture [19]

Fast-SCNN consists of four main modules: Learning to Downsample, Global Feature Extractor, Feature Fusion Module, and Classifier. The network first reduces the spatial dimensions of the input image while increasing feature richness, then extracts global context features and fuses them with fine spatial details. This fused representation is finally classified at the pixel level to

distinguish road and non-road areas. In the Icar system, this segmented output supports safe navigation by identifying drivable regions based on real-time camera input [19].

2.10 Iterative Closest Point (ICP)

Iterative Closest Point (ICP) is a widely used algorithm in known-map localization that aligns two 3D point clouds: one from the robot’s current observation and one from a reference map [27]. The output is a transformation matrix that minimizes the difference between the two point sets.

The ICP algorithm operates in three main steps: first, for each point in the input point cloud, it finds the nearest point in the reference point cloud (using nearest-neighbor search). Second, it calculates the centroids (center of mass) of both point clouds and aligns them via translation. Third, it applies rotation—typically using Singular Value Decomposition (SVD)—to best align the clouds.

These steps are repeated iteratively until convergence, defined by either reaching a maximum number of iterations or achieving a sufficiently low alignment error. The final result is the transformation matrix that aligns the current observation with the reference map, enabling accurate pose estimation.

2.11 Binary Robust invariant scalable keypoints (BRISK)

BRISK Binary Robust invariant scalable keypoints is a real-time feature method that combines a novel scale-space FAST detector with a binary bit-string descriptor built from intensity-comparison sampling. Keypoints are detected across scales via a FAST-based pyramid, then each is described by a 512-bit string derived from Gaussian-smoothed intensity comparisons on concentric sampling rings, with an orientation normalization step for rotation invariance. Evaluated on standard benchmarks, BRISK matches the accuracy of heavy float descriptors like SURF and SIFT while running up to an order of magnitude faster, making it highly suited for resource-constrained, real-time vision applications [14].

CHAPTER 3

FULL SYSTEM OVERVIEW

3.1 Overview of the Previous Icar System

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3.2 Overview of the Proposed System

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CHAPTER 4

LOW LEVEL SYSTEM

4.1 Electrical System

The Electrical System of Icar consists of two main subsystems: the Sensor System and the Actuation System. The Sensor System is responsible for gathering data from the environment and the vehicle itself, while the Actuation System controls the movement and operation of the vehicle based on the calculated data.

4.1.1 Sensor System

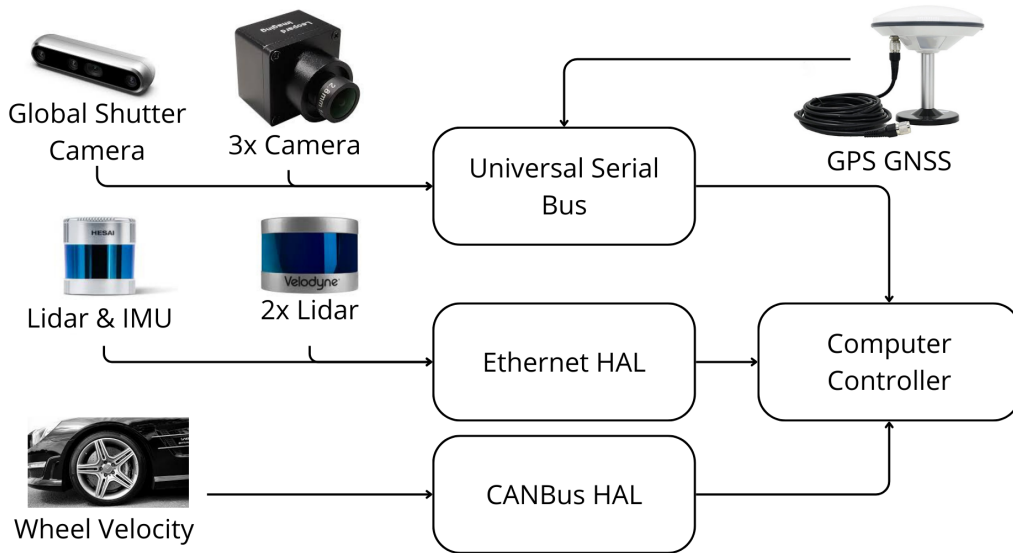


Figure 4.1. Block Diagram of Sensor System

Based on Figure 4.1, the Sensor System consists of several components, including LIDAR, Camera, GPS, and Wheel Velocity Sensor. The LIDAR sensors produce point cloud data and IMU data that interface with the ethernet controller of the computer. The Camera sensors produce image data that will be used for road segmentation, interfaced using the USB controller of the computer. The GPS sensor produces the estimation of latitude and longitude for the car that interfaces using USB. The Wheel Velocity Sensor is captured by sniffing the car's

main CAN bus using CANbus HAL module.

4.1.2 Actuation System

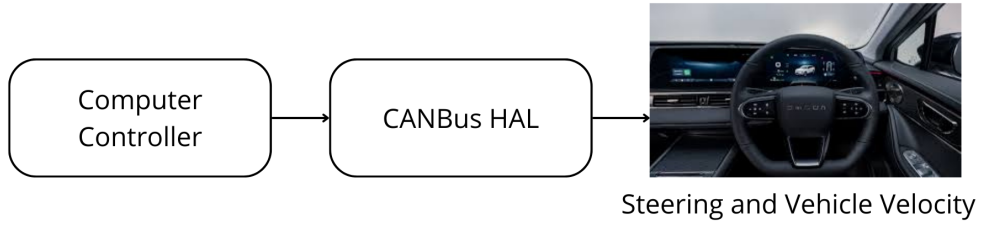


Figure 4.2. Block Diagram of Actuation System

Based on Figure 4.2, the Actuation System only proceeds Steering Angle and Velocity of vehicle. The commands are sent from the car's main CAN bus using CANbus HAL module. That technique is called Drive by Wire (DBW), where there is no physical connection between the driver and the actuator. The commands are generated using electrical signals from the computer to the car's main CAN bus.

4.2 Limitations

The limitations of the system are divided into two main categories: Sensor Limitations and Actuation Limitations. Each category has its own set of challenges that need to be addressed to ensure optimal performance of the autonomous vehicle.

4.2.1 Sensor Limitations

There are several limitations associated with the sensors used in the system. The limitations listed below:

- The noise measurement of GPS that can reach up to 2.5 meters in open sky condition, and can be worse in urban area with tall buildings or trees that can block the satellite signal.
- LIDAR Point Cloud has update rate limitation from 5 Hz to 20 HZ, which may affect the real-time performance for the higher level of system.
- There is 30 cm blindspot in front of car because of Lidar and Camera Mounting.

4.2.2 Actuation Limitations

There are several limitations associated with the actuation system used in the vehicle. The limitations listed below:

- The maximum car acceleration is 3.0 m/s^2 based on ISO 11270, which limits the speed of response for velocity control.
- The maximum steering angle rate is 5 deg/s based on car's specification, which limits the speed of response for steering control.
- The maximum steering angle position is $\pm 300 \text{ deg}$ based on car's specification, which limits the turning radius of the vehicle.

4.3 Low Level Program Architecture

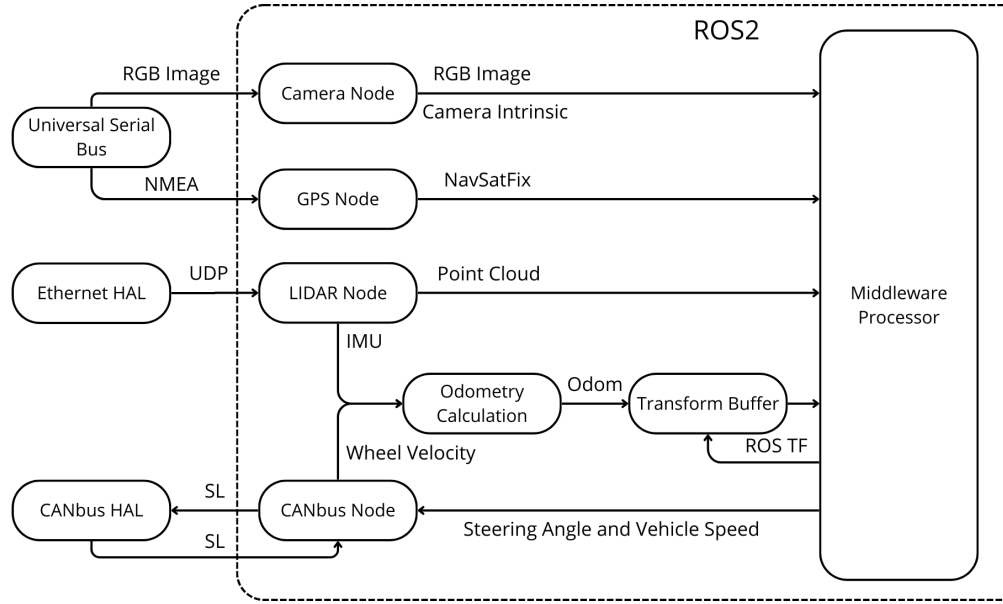


Figure 4.3. Block Diagram of Low Level Architecture

Based on Figure 4.3, the Low Level Program Architecture utilizes ROS2 as the framework for all processes. There are three main parts of Low Level Architecture, which are Sensor Data Acquisition, Odometry Calculation, and Frame Transformation.

4.3.1 Sensor Data Acquisition

Based on Figure 4.3, the Sensor Data Acquisition block is responsible for acquiring raw data from the LIDAR, camera, GPS, and wheel velocity sensor. For each sensor, we have a single independent ROS 2 node that publishes a standard ROS 2 message type. We use several standard ROS 2 message types in this part, such as `sensor_msgs/PointCloud2` for LIDAR point cloud data, `sensor_msgs/Image` for camera image data, `sensor_msgs/NavSatFix` for GPS data, and `std_msgs/Float32` for wheel velocity data.

4.3.2 Odometry Calculation

Based on Figure 4.3, the next step after Sensor Data Acquisition is Odometry Calculation. This block is responsible for computing the odometry from the wheel velocity sensor and the IMU data from the LIDAR. The

odometry is calculated using a differential-drive kinematic model, which takes into account the wheel velocities and the time interval between measurements. The resulting odometry data is then published as a `nav_msgs/Odometry` message type.

4.3.3 Frame Transformation

Based on Figure 4.3, there is a step called the transformation buffer. This part is responsible for managing the transformations between each sensor frame and the main frames, such as `base_link`, `odom`, and `map`. The diagram of the transformations between each frame is shown in Figure 4.4.

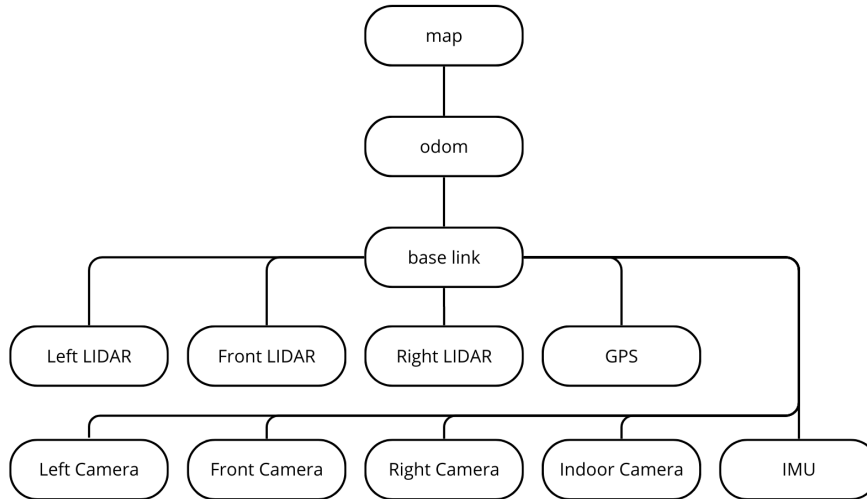


Figure 4.4. Transformation buffer

Figure 4.4 shows the transformations between each frame. The `base_link` frame is the main frame of the vehicle, which is located at the center of the vehicle. The `odom` frame represents the odometry, which is calculated from the wheel velocity sensor and IMU data from the LIDAR. The `map` frame is the global world frame that represents the global position of the vehicle. The transformations between frames are managed using the `tf2` library in ROS 2.

CHAPTER 5

MIDDLEWARE SYSTEM

5.1 Middleware System

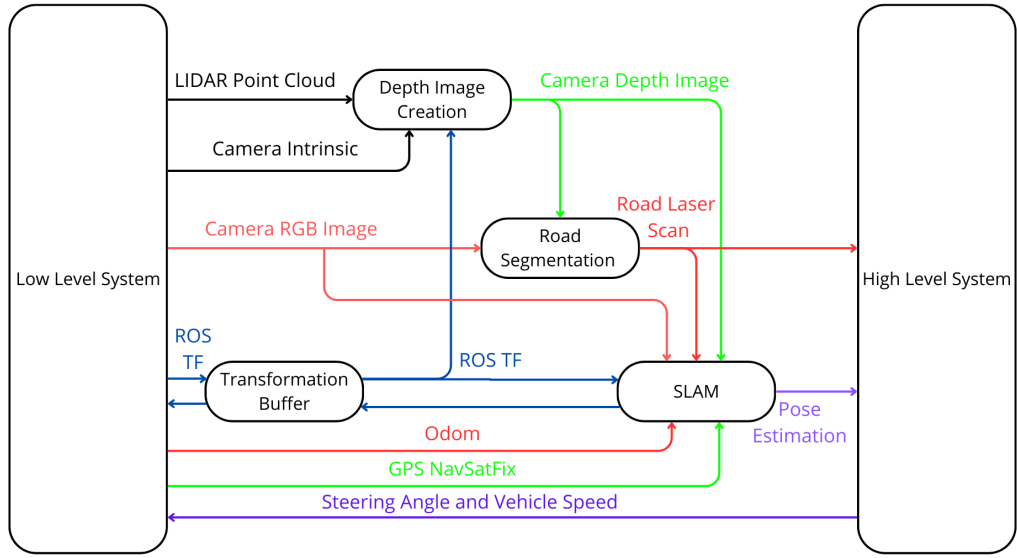


Figure 5.1. Block Diagram of Middleware System

Figure 5.1 shows the Middleware System that connects the Low Level Program Architecture with the High Level Program Architecture. There are three main parts in Middleware System, the first part is the Depth Image Creation, the second part is Road Segmentation using Machine Learning, and the third part is Graph Based SLAM for Pose Correction and Estimation.

5.2 Depth Image Creation

Base on Figure 5.1, the Depth Image Creation block is responsible for creating a depth image from the LIDAR point cloud data and camera intrinsic data. This process involves two main steps: LIDAR Concatenation and Post Filter for Depth Image.

5.2.1 LIDAR Concatenation

LIDAR Concatenation is the process of combining multiple LIDAR scans into a single point cloud. This is done to increase the density of the point cloud and improve the accuracy of the depth image. The concatenated point cloud is then projected onto the camera optical frame. The transformation lookup between the LIDAR frame and the camera frame can be obtained from the transformation buffer in the Low Level Program Architecture 4.4.

After transformed point cloud to the camera optical frame, we can calculate the depth image using the camera intrinsic parameters. Each point in the point cloud is projected onto the image plane using the pinhole camera model, and the depth value is calculated based on the distance from the camera to the point.

5.2.2 Post Filter for Depth Image

The raw depth image obtained from the previous step has some hole areas due to the sparse nature of the LIDAR point cloud. To address this issue, a post-filtering process is applied to fill in the holes and smooth the depth image. This process involves using morphological operations to fill in the holes by neighboring pixel values.

5.3 Machine Learning

Based on figure 5.1, another sub process of middleware is Road Segmentation. Road Segmentation is done using Machine Learning approach. The idea is to segment each pixel of the image into two classes, which are road and non-road. The segmentation result will be transformed base link frame using depth image from previous step. The output of this process is a laser scan msg that will be used by high level system to determine target velocity and steering angle.

5.3.1 Architecture Design

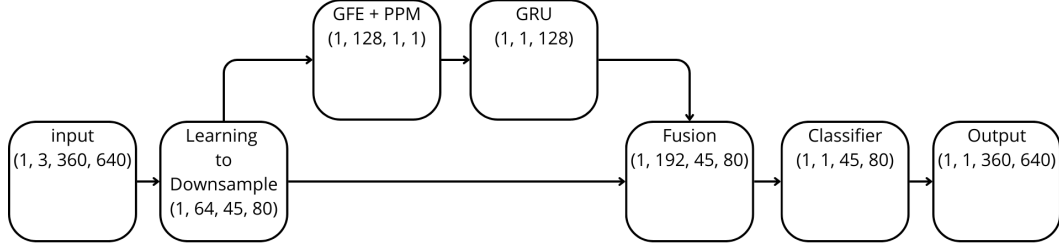


Figure 5.2. Machine Learning Architecture

Figure 5.2 visualize the architecture of our proposed new Machine Learning method. Based on Fast-SCNN, we proposed our new improvement method by adding GRU (Gated Recurrent Unit). The main idea of adding GRU to the existence of Fast-SCNN is to make a global context. That method can make the temporal effect on detection.

5.3.2 Dataset Preparation

The preparation of dataset is crucial to Machine Learning process. The dataset only have one class called road class. The dataset image captured by driving the car through various road conditions, such as different lighting conditions, weather conditions, and road types. The dataset is then annotated manually to create ground truth labels for the road class.

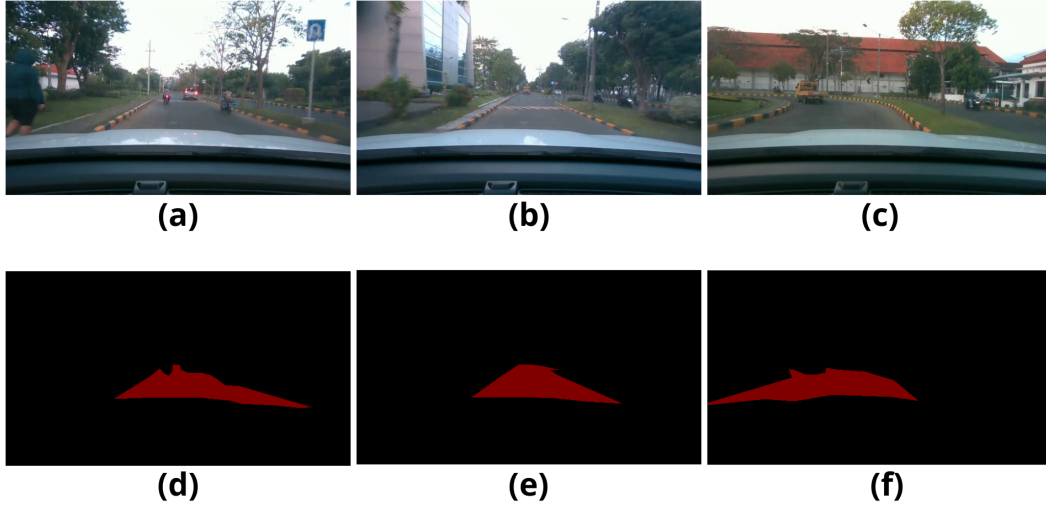


Figure 5.3. DATasets for Machine Learning

Figure 5.3 shows some example of 1000 images dataset that we used for training and validation. The image a, b, and c are the rgb input image, while the image d, e, and f are the corresponding ground truth labels for road class. The red segmented area in the ground truth labels represents the road class.

5.3.3 Model Training

The model training process is carried out using the prepared dataset. The problem of our dataset is imbalanced data for example on figure 5.3 in sub image b, there is a road bump that have same color and pattern with road barrier but we want to predict that it's a road too. To handle that imbalanced data we use the Tversky loss function during training. The Tversky loss function can handle imbalanced data and bunch of false positive cases [1]. The important of the bunch of false positive cases is to avoid the vehicle from misdetecting the road area as non-road area, which can lead to unsafe driving conditions.

Another technique that we use to improve the model performance is data augmentation. Data augmentation is done by applying various transformations to the input images, such as random gamma, random brightness, CLAHE, HSV limit, RGB shift, random shadow, random sunflare, ISO noise, Motion blurr,

random tone curve, and random brightness contrast. These transformations help to increase the diversity of the training data and improve the model's ability to generalize to unseen data.

Another important aspect of the model training process is hyperparameter tuning. We experiment with different learning rates, batch sizes, and number of epochs to find the optimal combination that yields the best performance on the validation set. The model is trained using the Adam optimizer with an initial learning rate of 0.001 and a batch size of 12. The training is carried out for 200 epochs, with early stopping implemented to prevent overfitting.

Figure 5.2 shows that there's a GRU block after the feature extractor block. In the training process we disable saving the global context because our dataset were very random and not sorted yet.

5.3.4 Model Inference

After training process has done, the model was saved in pth format. That model deployed on Icar's PC that have embedded nvidia GPU with 8Gb of VRAM. Because the model is very lightweight, we don't need to convert it to onnx or another optimizer. We just load the model in pth format and doing computation in GPU.

5.3.5 Post Processing

There is an important step of Road Segmentation called Post Processing. There are step by step that we have done in post processing. The first step is CLAHE process, the CLAHE (Contrast Limited Adaptive Histogram Equalization) process is done to improve the contrast of the image. The seconds step is ROI crop that will erase all unused pixel that will not be the road. The third step is morphological operation, the morphological operation is done to remove small noise and fill small holes in the segmented image. The last step is depth image transformation, the depth image transformation is done to transform the segmented image from camera frame to base link frame using depth image from previous step. The fourth step is largest contour detection, the largest contour detection is done to find the largest road area

in the segmented image. The fifth step is doing a low pass filter between time steps for every pixel to make the segmentation result more stable.

5.3.6 Final Transformation

The final step of Road Segmentation is Final Transformation. The final transformation is done by using the calculated depth image from 5.2 to transform the segmented image from camera frame to base link frame. The output of this process is a laser scan msg that will be used by high level system to determine target velocity and steering angle.

The first process of final transformation is doing a high pass filter on segmented area. The high pass filter is done to get the edge of the road area. After that, we do a laser scan for 0 degree to 180 degree in base link frame with 0.01 degree resolution. For every angle, we search for the closest pixel that has segmented area in the depth image. The distance from base link frame to that pixel is the range value for that angle in laser scan msg.

5.4 Graph Based SLAM

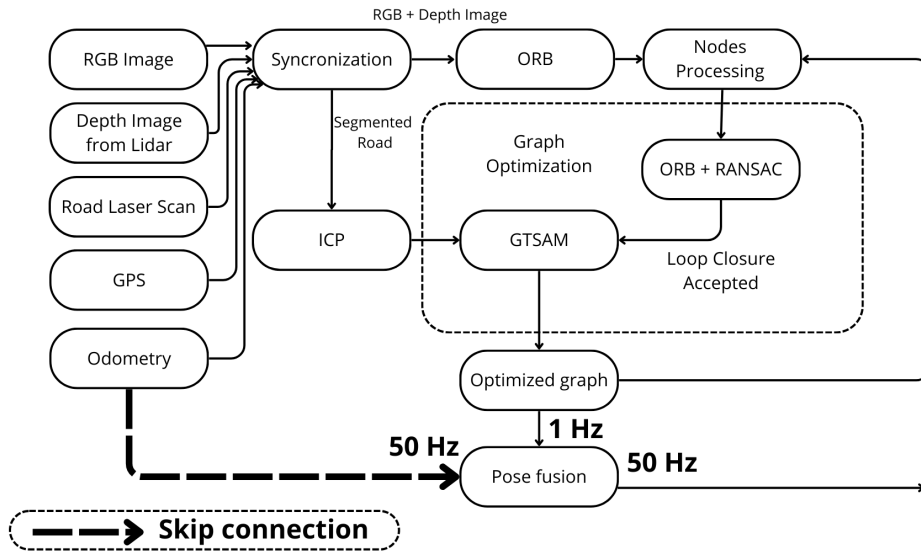


Figure 5.4. Block Diagram of Graph Based SLAM based on Extended RTAB-Map [15]

Figure 5.4 is the internal explanation of SLAM block in figure 5.1. The Graph Based SLAM is responsible for providing the best estimation of vehicle pose by fusing all sensor data from RGB image, Depth image, Road laser scan, GPS, and Odometry from wheel encoder and IMU. The Graph Based SLAM consists of several main parts, which are Synchronization, Nodes Processing, Loop Closure Detection, Graph Optimization, and Pose Fusion.

The difference between our improved Graph Based SLAM with the previous Extended RTAB-Map [15] is the addition of GPS sensor to be a first guess for loop closure finding process. Another difference is doing ORB feature extraction instead of using GFTT because [9] said that ORB better than GFTT in outdoor environment.

5.4.1 Graph Based SLAM Mode

The Graph Based SLAM has two modes of operation, which are Mapping Mode and Localization Mode. In Mapping Mode, the Graph Based SLAM builds a map of the environment while simultaneously estimating the vehicle's pose. This mode is typically used during the initial exploration of an unknown environment. In Localization Mode, the Graph Based SLAM uses the previously built map to estimate the vehicle's pose without updating the map. This mode is typically used when the vehicle is operating in a known environment.

5.4.2 Synchronization

The first step of Graph Based SLAM is Synchronization. Based on [15], the synchronization strategy has been done using approximate time synchronization. The approximate time synchronization is done to synchronize all sensor data from RGB image, Depth image, Road laser scan, GPS, and Odometry from wheel encoder and IMU. The synchronized data is then used for the next step,

5.4.3 Nodes Processing

The first step of Nodes Processing is feature extraction of the image from camera. The feature extraction is done using ORB (Oriented FAST

and Rotated BRIEF) [23] to extract keypoints and descriptors from the RGB image and Depth image. After the extraction done, the Node controller will check if the current node is new or revisited node. If the current node is a new node, the node controller will add the new node to the graph. The new node will contain the RGB image, Depth image, Road laser scan, GPS, and Odometry data. If the current node is a revisited node, the node controller will proceed to the Loop Closure Detection step.

5.4.4 Loop Closure Detection

After all the feature extraction has done, the next step is to find loop closure candidates. By doing first guess using GPS data (LoopGPS=true), the loop closure will only search for candidate inside LocalRadius searching area [12]. Like the method from [15], the loop closure detection is done using feature matching between the current node and the nodes inside the LocalRadius searching area. The feature matching is done using ORB descriptors and RANSAC to filter out outliers. If the number of inliers is greater than a predefined threshold, the loop closure hypothesis is added.

5.4.5 Graph Optimization

After the loop closure hypothesis has been found, the next step is Graph Optimization. The graph optimization is done using GTSAM (Georgia Tech Smoothing and Mapping) [4] to optimize the graph and find the best estimation of vehicle pose by using the ICP constraint and odometry constraint. The graph optimization is done using the Levenberg-Marquardt algorithm to minimize the error between the predicted pose and the observed pose from the sensor data. The optimized graph is then used for the next step, which is Pose Fusion.

5.4.6 Pose Fusion

Like the method from [15], the Pose Fusion is done using an Extended Kalman Filter (EKF) to fuse the optimized pose from the Graph Optimization step with the odometry data from the wheel encoder and IMU. The fusion strategy use an odometry data as a velocity by differentiating the position

data from odometry. The pose estimation from Graph Optimization is used as a measurement for position update in EKF. The output of the Pose Fusion is the best estimation of vehicle pose, which is then used by the High Level Program Architecture for trajectory planning and control.

CHAPTER 6

HIGH LEVEL SYSTEM

6.1 High Level Program Architecture

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6.1.1 Graph Based SLAM and Machine Learning Integration

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6.1.2 Model Prediction Control

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6.1.3 Trajectory Optimization

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6.1.4 Target Velocity and Steering Calculation

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6.2 Safety System

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6.2.1 ISO 26262 Implementation

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6.2.2 Emergency Stop System

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6.2.3 Safety Tests

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CHAPTER 7

RESULT AND DISCUSSION

7.1 Result of GPS Only vs Graph Based SLAM

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7.1.1 Pose Estimation Accuracy

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7.1.2 Pose Estimation Precision

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7.3 Previous Navigation System vs Proposed Navigation System

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7.3.1 Trajectory Generation

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7.3.2 Safety System

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7.3.3 The Important of Prediction

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7.4 Ablation Studies

7.4.1 Ablation Study on Road Segmentation

7.4.2 Ablation Study on Pose Estimation

CHAPTER 8

CONCLUSION

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List of Publications

1. Muhtadin, A. W. Maulana, R. Dikairono and A. Zaini, "Omnivision Calibration on Mobile Robot Using Machine Learning," *2024 International Conference on Computer Engineering, Network, and Intelligent Multimedia (CENIM)*, Surabaya, Indonesia, 2024, pp. 1–8, doi: 10.1109/CENIM64038.2024.10882712.

Research History

1. Multicast Communication System on IRIS Robot 2022
2. Natural Ball Handling System on IRIS Robot 2022
3. Omnivision Camera Calibration on Mobile Robot Using Machine Learning 2023
4. Robot Positioning System Using Gaussian Model on IRIS Robot 2023
5. Empty Goal Detection System Using Omnivision Camera on IRIS Robot 2023
6. Content Management System and User Interface for RAISA ITS Robot 2024
7. Custom Operating System for Autonomous Vehicle and IRIS Robot 2024
8. Empty Goal Detection System Using Stereo Depth Camera on IRIS Robot 2024
9. Ball Detection System Using Spatial Image Classification on IRIS Robot 2024
10. Hardware Communication System Using Layer 2 Communication for Autonomous Vehicle and IRIS Robot 2024
11. Hardware Communication System Using CAN-bus for Autonomous Vehicle and IRIS Robot 2024
12. End to End Machine Learning System for Autonomous Vehicle 2025
13. Fleet Management System for AMR (Autonomous Mobile Robot) 2025

Other Background

Before venturing into the world of robotics and programming, the author was involved in music. As a musician, he mastered various musical instruments such as guitar, bass, and keyboard. In addition to playing instruments, he was also active in the field of mixing and mastering. His hobby of signal processing sparked an interest in learning how computers process signals. This passion led him to pursue a degree in Computer Engineering at ITS. Moreover, he delved into robotics to understand how computers and hardware operate, enabling him to perform advanced digital signal processing techniques.